

Rheology of concentrated suspensions of deformable elastic particles such as human erythrocytes

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Abstract

New models for the viscosity of concentrated suspensions of deformable elastic particles are developed using the differential effective medium approach (DEMA). The models are capable of describing the rheological behavior of un-aggregated suspensions of human red blood cells (RBCs). With the increase in shear rate, a shear-thinning behavior is predicted similar to that observed in the case of un-aggregated suspensions of RBCs. A decrease in relative viscosity and an enhancement of shear-thinning behavior is predicted when either the particle rigidity (elastic modulus) is decreased or the continuous medium viscosity is increased. These predictions are similar to those observed in suspensions of human RBCs. The proposed models are evaluated using experimental data on normal and hardened human RBC suspensions in protein-free saline.

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1. Introduction

Blood is a suspension of red blood cells (erythrocytes) (RBCs), white blood cells (leukocytes), and platelets in plasma. RBCs are biconcave discs about 8 μm in diameter (Skalak et al., 1989) and 1 μm in thickness at the center (maximum thickness is about 2.5 μm). The RBCs consist of a thin elastic membrane filled with a hemoglobin solution that is a Newtonian fluid with viscosity of about 6–7 mPa s (Snabre and Mills, 1999). RBCs are readily deformable, a characteristic that allows them to pass through the smallest capillaries; they are manufactured in the bone marrow and their primary function is the transport of oxygen. The white blood cells are generally spherical with a mean diameter of about 7 μm . They are much less deformable than the RBCs and their primary function is in the body's defenses and immunity reactions. Platelets are irregularly oval or spherical bodies without any nucleus. They are usually 2–4 μm in their largest dimension and they play an important role in blood clotting and sealing of an injured blood vessel. Blood plasma, the continuous

medium of blood, is about 90% water and contains a number of proteins that make up about 7% of its weight. Blood plasma is a Newtonian fluid with viscosity of about 1.2 mPa s at 37°C (Chien et al., 1966; Levick, 2000; Skalak et al., 1989).

The rheological properties of the blood play an important role in the regulation of blood flow in micro and macrovessels. There is a growing body of evidence linking blood rheology to commonly accepted risk factors of atherosclerosis (heart disease and stroke) such as: smoking, cholesterol, diabetes, and gender.

Blood is known to be a non-Newtonian fluid with non-linear stress–strain rate relationship (Meiselman, 1965; Cheng and Deville, 1996; van de Vosse and Gijsen, 1998; Marinakis and Tsangaris, 1998; Gijsen et al., 1999a,b; Zhang and Kuang, 2000; Kim et al., 2000; Kuke et al., 2001; Liepsch, 2000; McMillan, 1995a,b; Liepsch et al., 1991, 1995; Hosoda et al., 1994; Gahtgens, 1980; Gahtgens et al., 1980; Gahtgens and Schmid-Schonbein, 1982). As the RBCs constitute more than 99% of the particulate matter in blood, they govern the flow properties of blood. The white blood cells and platelets usually occupy only less than 1% of the total volume of blood cells in normal human blood

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and therefore, they exert little influence on the bulk rheology of blood. The volume fraction of RBCs in blood is called the “hematocrit”; the normal range is 0.47 ± 0.07 in adult human males and 0.42 ± 0.05 in adult females. As the blood rheology is governed by RBCs, several experimental studies have been reported on the viscosity of RBC suspensions (Taylor et al., 1965; Chien et al., 1966, 1970; Brooks et al., 1970; Chien, 1970; Brennen, 1975; Thurston, 1979; Carr and Cokelet, 1981; Kim et al., 2000). The viscosity of RBC suspensions is strongly influenced by the following three factors: (a) volume fraction of RBC—with the increase in the volume fraction of RBC, the suspension viscosity increases; (b) cell deformability—the suspension viscosity increases with the decrease in RBC deformability (Schmid-Schonbein et al., 1969; Chien, 1977); and (c) aggregation of RBCs—RBCs tend to pile up into rod-shaped aggregates, like a pile of coins, when at rest (Schmid-Schonbein et al., 1973; Armstrong et al., 1997, 1999; Baskurt and Meiselman, 1997a,b; Baskurt et al., 2002). Such piles of cells are called “rouleaux”. Rouleaux formation is believed to be due to bridging between the cells by plasma proteins such as fibrinogen (Merill et al., 1978). Aggregation of RBCs results in an increase in the viscosity of blood.

In order to consider just the effects of hematocrit (volume fraction of RBCs) and RBC deformability on blood rheology, researchers have studied the rheological properties of RBCs suspended in media where rouleaux formation does not occur; one such medium is isotonic saline (Ringer solution). Chien and co-workers (Chien et al., 1970; Chien, 1977) studied the effect of cell deformability on the relative viscosity (ratio of suspension viscosity to continuous phase viscosity) of unaggregated RBC suspensions. Human erythrocytes were

hardened to different degrees using acetaldehyde. Fig. 1 shows the general behavior of RBC suspensions observed by Chien and co-workers for a fixed volume fraction of RBC. The suspension of normal RBCs exhibits shear-thinning behavior due to deformation of cells. A progressive shift of relative viscosity (η_r)–stress curve to the right occurs as the RBCs gradually lose their deformability during slow fixation in acetaldehyde solution. The suspension of rigid RBCs exhibit Newtonian behavior and has the highest viscosity.

In shear flow, an erythrocyte can exhibit two types of motion (Fischer and Schmid-Schonbein, 1977; Fischer et al., 1978a,b; Keller and Skalak, 1982; Barthes-Biesel and Sgaier, 1985; Bitbol, 1986): *flipping/tumbling* motion in which the shape of the cell remains close to its original resting shape (biconcave disc), and *tank-treading* motion in which the cell deforms into an ellipsoid-like particle at a steady orientation with the membrane rotating about the cell interior similar to the motion of a tank tread. When the RBC undergoes flipping/tumbling motion, it behaves more like a flexible/deformable elastic solid. The behavior of RBC undergoing tank-treading motion, however, is similar to that of a liquid droplet (Schmid-Schonbein and Wells, 1969; Schmid-Schonbein et al., 1971); the transmission of stresses across the cell membrane takes place and the cytoplasm of the RBC participates in shear flow. Keller and Skalak (1982) have shown that the transition from a flipping motion to a tank-treading motion of the RBC depends on the erythrocyte elongation and the ratio of the inner to outer liquid viscosities. Only when the viscosity ratio is small and the shear rates are high, a tank-tread motion of the RBC membrane is predicted. According to Barthes-Biesel and Sgaier (1985) also, the RBCs behave as flexible elastic solids, and have a flipping motion even at very high shear rates; the tank-treading motion is expected only when the suspending medium of the RBCs has a high viscosity so that the viscosity ratio (inner to outer liquid viscosities) is small. In a normal human blood, the viscosity ratio is about 6, high enough to suppress the tank-treading motion. Thus, it is quite appropriate to treat RBCs as deformable elastic particles in modelling the rheological behavior of RBC suspensions.

A vast amount of published literature exists on the rheology of suspensions of rigid particles (Einstein, 1906, 1911; Mooney, 1951; Roscoe, 1952; Brinkman, 1952; Krieger and Dougherty, 1959; Krieger, 1972; Brenner, 1972; Jeffery and Acrivos, 1976; Batchelor, 1977; Macosko, 1994; Larson, 1999). However, suspensions of deformable elastic particles have received less attention. In this paper, new equations for the viscosity of concentrated suspensions of deformable elastic particles are developed. The equations developed are compared with experimental data on suspensions of human RBCs.

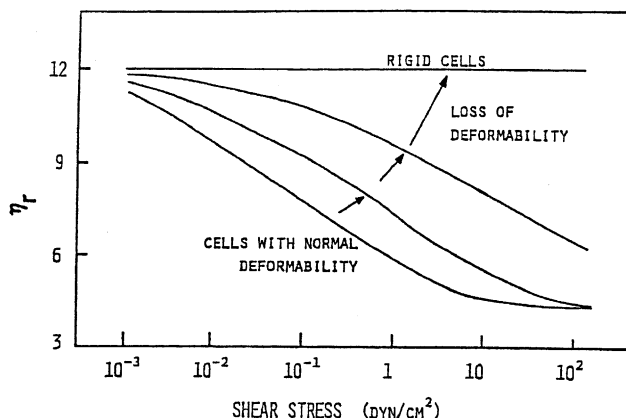


Fig. 1. Relative viscosity (η_r) as a function of shear stress for suspensions of human red cells in Ringer's solution (Chien, 1977); the arrows show the progressive shift of the curves towards higher viscosities as the red cells gradually lose their deformability during slow fixation in acetaldehyde solution.

2. Development of new viscosity equations

New equations for the viscosity of concentrated suspensions of deformable elastic particles are derived using the differential effective medium approach (DEMA). According to this approach, a concentrated suspension is considered to be obtained from an initial continuous phase by successively adding infinitesimally small quantities of particles to the system while the final volume fraction of the dispersed phase is reached. At any arbitrary stage (i) of the process, the addition of an infinitesimal amount of particles leads to the next stage ($i + 1$). The suspension of stage (i) is then treated as an equivalent “effective medium”, which is homogeneous with respect to the new set of particles added to reach stage ($i + 1$). The solution of a dilute suspension is then applied to determine the increment increase in viscosity in going from stage (i) to stage ($i + 1$). The differential equation derived in this manner is integrated to obtain the final solution for a concentrated suspension. This DEMA was originally proposed by Bruggeman (1935), who derived an equation for the dielectric constant of a concentrated suspension. The DEMA is often referred to as “Brinkman–Roscoe” differential scheme in the rheology literature as Brinkman (1952) and Roscoe (1952) were the first to apply this scheme for the derivation of a viscosity equation for concentrated suspensions of *rigid* spherical particles. Brinkman (1952) and Roscoe (1952) utilized the celebrated Einstein viscosity equation (Einstein, 1906, 1911) as a solution of dilute suspension in the derivation of their viscosity equation for concentrated suspensions.

In the present work, the DEMA is applied to derive the viscosity equations for concentrated suspensions of deformable elastic particles. The solution for a dilute suspension of deformable elastic particles is given by Goddard and Miller (1967) as

$$\eta_r = 1 + \phi \left(\frac{[\eta](1 - \frac{3}{2}N_{se}^2)}{1 + (\frac{3}{2}N_{se})^2} \right), \quad (1)$$

where η_r is the relative viscosity, defined as the ratio of suspension viscosity (η) to continuous medium viscosity (η_c), ϕ is the volume fraction of the dispersed-phase, $[\eta]$ is the intrinsic viscosity (equal to 2.5 for rigid spherical particles), N_{se} is the shearoelastic number given by

$$N_{se} = \frac{\eta_c \dot{\gamma}}{G_p}. \quad (2)$$

In Eq. (2), $\dot{\gamma}$ is the imposed shear rate, and G_p is the shear modulus of elastic particles. Eq. (1) is valid only for dilute suspensions; it cannot be applied to concentrated suspensions as hydrodynamic interactions between the particles are ignored in its derivation.

Let us now consider a suspension of deformable elastic particles with a volume fraction of particles ϕ and viscosity $\eta(\phi)$. Into this suspension, an infinitesimally

small amount of new particles is added. The increment increase in viscosity $d\eta$ resulting from the addition of the new particles can be calculated from the Goddard and Miller equation (Eq. (1)) by treating the suspension into which the new particles are added as an equivalent homogeneous medium of viscosity $\eta(\phi)$. Thus,

$$d\eta = \eta \left[\frac{[\eta] \left\{ 1 - \frac{3}{2}(\eta \dot{\gamma} / G_p)^2 \right\}}{1 + (\frac{3}{2}(\eta \dot{\gamma} / G_p))^2} \right] d\phi. \quad (3)$$

This equation can be re-written as

$$\frac{d\eta}{\eta \left[1 - \frac{3}{2}(\eta \dot{\gamma} / G_p)^2 \right]} + \frac{(\frac{3}{2}(\eta \dot{\gamma} / G_p))^2}{\eta \left[1 - \frac{3}{2}(\eta \dot{\gamma} / G_p)^2 \right]} d\eta = [\eta] d\phi. \quad (4)$$

Eq. (4) can be further re-arranged as

$$\left[\frac{1}{\eta} + \frac{a\eta}{1 - a\eta^2} + \frac{3}{2}a\eta + \frac{3}{2}a^2 \left(\frac{\eta^3}{1 - a\eta^2} \right) \right] d\eta = [\eta] d\phi, \quad (5)$$

where $a = (3/2)(\dot{\gamma} / G_p)^2$. Upon integration with the limit $\eta \rightarrow \eta_c$ at $\phi \rightarrow 0$, Eq. (5) gives

$$\eta_r \left[\frac{1 - \frac{3}{2}N_{se}^2\eta_r^2}{1 - \frac{3}{2}N_{se}^2} \right]^{-5/4} = \exp([\eta]\phi). \quad (6)$$

Eq. (6), referred to as model 1 in the remainder of the paper, reduces to the following equation in the limit $N_{se} \rightarrow 0$:

$$\eta_r = \exp([\eta]\phi). \quad (7)$$

Model 1 is expected to describe the viscosity of suspensions at low values of ϕ ($\phi < 0.2$). At high ϕ , model 1 is expected to underpredict η_r . This is because in the derivation of the differential equation (Eq. (3)) leading to model 1 (Eq. (6)) it is assumed that all the volume of the suspension before new particles are added is available as free volume to the new particles. In reality, the free volume available to disperse the new particles is significantly less, due to the volume pre-empted by the particles already present. The increase in the actual volume fraction of particles when new particles are added to the suspension is $d\phi/(1 - \phi)$. Thus,

$$d\eta = \eta \left[\frac{[\eta] \left\{ 1 - \frac{3}{2}(\eta \dot{\gamma} / G_p)^2 \right\}}{1 + (\frac{3}{2}(\eta \dot{\gamma} / G_p))^2} \right] \frac{d\phi}{1 - \phi}. \quad (8)$$

Eq. (8) can be re-written as

$$\left[\frac{1}{\eta} + \frac{a\eta}{1 - a\eta^2} + \frac{3}{2}a\eta + \frac{3}{2}a^2 \left(\frac{\eta^3}{1 - a\eta^2} \right) \right] d\eta = [\eta] \frac{d\phi}{1 - \phi}, \quad (9)$$

where $a = (3/2)(\dot{\gamma} / G_p)^2$. Upon integration with the limit $\eta \rightarrow \eta_c$ at $\phi \rightarrow 0$, Eq. (9) gives

$$\eta_r \left[\frac{1 - \frac{3}{2}\eta_r^2 N_{se}^2}{1 - \frac{3}{2}N_{se}^2} \right]^{-5/4} = (1 - \phi)^{-[\eta]}. \quad (10)$$

This equation, referred to as model 2 in the remainder of the paper, reduces to the following equation in the limit $N_{se} \rightarrow 0$:

$$\eta_r = (1 - \phi)^{-[\eta]} \quad (11)$$

One limitation of models 1 and 2 is that they fail to account for the so-called “crowding effect” caused by packing difficulty of particles at high ϕ (Mooney, 1951; Krieger and Dougherty, 1959). Due to the immobilization of some of the continuous-phase fluid in the voids between the existing particles, the free volume of the continuous-phase fluid available when new particles are added is significantly less than $(1 - \phi)$. Accordingly, Mooney (1951), in the derivation of his well-known equation for the viscosity of concentrated suspensions of rigid particles, contended that the incremental increase in the volume fraction of the dispersed phase when infinitesimal amount of new particles are added to an existing suspension of dispersed phase volume fraction ϕ , is $d[\phi/(1 - \phi/\phi_m)]$ rather than $d\phi/(1 - \phi)$ as used in the derivation of model 2 (Eq. (10)). Note that ϕ_m is the maximum packing volume fraction of particles; for random close packing of monosized spherical particles, ϕ_m is 0.637. Thus, Eq. (3) becomes

$$d\eta = \eta \left[\frac{[\eta] \{1 - \frac{3}{2}(\dot{\gamma}/G_p)^2\}}{1 + (\frac{3}{2}(\dot{\gamma}/G_p)^2)} \right] d \left[\frac{\phi}{1 - (\phi/\phi_m)} \right]. \quad (12)$$

Eq. (12) can be re-written as

$$\left[\frac{1}{\eta} + \frac{a\eta}{1 - a\eta^2} + \frac{3}{2}a\eta + \frac{3}{2}a^2 \left(\frac{\eta^3}{1 - a\eta^2} \right) \right] d\eta = [\eta] d \left[\frac{\phi}{1 - (\phi/\phi_m)} \right], \quad (13)$$

where $a = (3/2)(\dot{\gamma}/G_p)^2$. Upon integration with the limit $\eta \rightarrow \eta_c$ at $\phi \rightarrow 0$, Eq. (13) gives

$$\eta_r \left[\frac{1 - \frac{3}{2}\eta_r^2 N_{se}^2}{1 - \frac{3}{2}N_{se}^2} \right]^{-5/4} = \exp \left[\frac{[\eta]\phi}{1 - (\phi/\phi_m)} \right]. \quad (14)$$

This equation, referred to as model 3 in the remainder of the paper, reduces to the following equation in the limit of $N_{se} \rightarrow 0$:

$$\eta_r = \exp \left[\frac{[\eta]\phi}{1 - (\phi/\phi_m)} \right]. \quad (15)$$

Krieger and Dougherty (1959), in the derivation of their well-known equation for the viscosity of concentrated suspensions of rigid particles, argued that the incremental increase in the volume fraction of the dispersed phase when a small amount of new particles are added to an existing suspension of concentration ϕ , is $d\phi/(1 - \phi/\phi_m)$ rather than $d[\phi/(1 - \phi/\phi_m)]$ as thought by

Mooney (1951). Hence, Eq. (3) becomes

$$d\eta = \eta \left[\frac{[\eta] \{1 - \frac{3}{2}(\dot{\gamma}/G_p)^2\}}{1 + (\frac{3}{2}(\dot{\gamma}/G_p)^2)} \right] \left(\frac{d\phi}{1 - (\phi/\phi_m)} \right). \quad (16)$$

Eq. (16) can be further re-arranged as

$$\left[\frac{1}{\eta} + \frac{a\eta}{1 - a\eta^2} + \frac{3}{2}a\eta + \frac{3}{2}a^2 \left(\frac{\eta^3}{1 - a\eta^2} \right) \right] d\eta = [\eta] \frac{d\phi}{(1 - (\phi/\phi_m))}, \quad (17)$$

where $a = (3/2)(\dot{\gamma}/G_p)^2$. Upon integration with the limit $\eta \rightarrow \eta_c$ at $\phi \rightarrow 0$, Eq. (17) gives

$$\eta_r \left[\frac{1 - \frac{3}{2}\eta_r^2 N_{se}^2}{1 - \frac{3}{2}N_{se}^2} \right]^{-5/4} = \left[1 - \frac{\phi}{\phi_m} \right]^{-[\eta]\phi_m}. \quad (18)$$

This equation, referred to as model 4 in the remainder of the paper, reduces to the following equation in the limit of $N_{se} \rightarrow 0$:

$$\eta_r = \left(1 - \frac{\phi}{\phi_m} \right)^{-[\eta]\phi_m}. \quad (19)$$

As ϕ_m , the maximum packing volume fraction of undeformed particles, is sensitive to the particle size distribution, models 3 and 4 are capable of taking into account the effect of the particle size distribution on the viscosity of suspensions. An increase in ϕ_m occurs when a monodisperse suspension is changed to a polydisperse suspension.

3. Predictions of new viscosity equations

Fig. 2 shows the relative viscosities predicted from models 1 to 4. All models predict that the relative viscosity, at any given value of particle volume fraction ϕ , decreases with the increase in the shear-elastic number N_{se} . This shear-thinning behavior is due to deformation of particles with the increase in N_{se} . With the increase in ϕ , the relative viscosity increases, as expected. The relative viscosity prediction of different models for a given ϕ are in the following order: η_r (model 3) > η_r (model 4) > η_r (model 2) > η_r (model 1). In the calculations shown in Fig. 2, $[\eta]$ was taken to be 2.5 and ϕ_m for models 3 and 4 was taken to be 0.637. ϕ_m of 0.637 corresponds to random close packing of uniform spheres. The calculations shown in Fig. 2 were restricted to $N_{se} < 1.0$ as the Goddard and Miller equation (Eq. (1)), and consequently the new models (models 1–4), are valid for $N_{se} < 1.0$.

Fig. 3 shows the effects of particle rigidity and continuous medium viscosity on the relative viscosity of a concentrated suspension ($\phi = 0.45$) of deformable elastic particles. The plots are generated from model 4

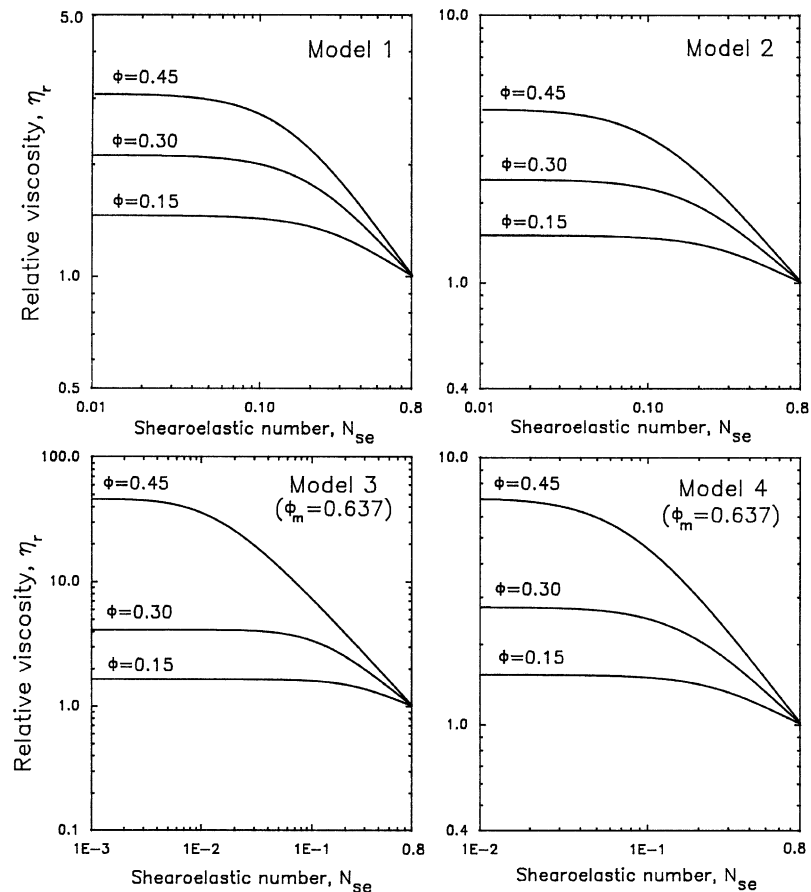


Fig. 2. Relative viscosity (η_r) as a function of shearoelastic number (N_{se}) predicted by the proposed models. The predictions are shown for various values of particle volume fraction (ϕ). The intrinsic viscosity $[\eta]$ is taken to be 2.5 in the calculations.

(Eq. (18)) with $[\eta] = 2.5$ and $\phi_m = 0.637$. In the top plot of Fig. 3, relative viscosity is shown as a function of shear stress ($= \eta_c \dot{\gamma}$) acting on the particles, for different values of G_p (particle shear modulus). The rigid particle suspension ($G_p \rightarrow \infty$) behaves as a Newtonian fluid; this behavior is the same as that exhibited by a suspension of hardened RBCs (see Fig. 1). With the decrease in G_p , shearing-thinning behavior is observed similar to that of RBC suspensions of Fig. 1. The bottom plot of Fig. 3 shows relative viscosity as a function of shear rate for different values of the continuous-medium viscosity (η_c). With the increase in η_c , the η_r -shear rate curve shifts to the left. This indicates that the shear-thinning behavior in suspension begins at a lower shear rate when η_c is increased. In the shear rate range where shear-thinning is present in the suspension, the relative viscosity decreases with the increase in η_c . A similar behavior is exhibited by suspensions of normal human RBC when the continuous-phase viscosity is increased (Chien, 1977). Note that a decrease in the relative viscosity and enhancement of shear-thinning behavior exhibited by the suspension, when G_p is decreased and η_c is increased, can be explained in terms of an increase in the deformation of particles.

4. Comparison of new viscosity equations with experimental data

Experimental viscosity data on suspensions of normal and hardened human erythrocytes are considered to evaluate the viscosity equations developed in the paper. The data are described only briefly here; further details about these data can be found in the original references. Chien et al. (1966) reported the results on the viscosity of normal human RBC suspensions in protein-free Ringer solution (saline). The suspension of RBCs in saline eliminates the problem of cell aggregation. The data were reported at four shear rates (52, 5.2, 0.52, and 0.052 s^{-1}) as a function of hematocrit (volume fraction of RBC). The viscosity measurements were made at 37°C . Chien et al. (1970) studied the rheological behavior of hardened human erythrocytes. The suspensions of acetaldehyde-fixed (hardened) RBCs in protein-free saline were studied using a coaxial cylinder viscometer at 37°C . The volume fraction of RBC was varied from 0 to 0.522. Brooks et al. (1970) measured the viscosity as a function of shear rate and volume concentration for normal and acetaldehyde-fixed human erythrocytes suspended in protein-free

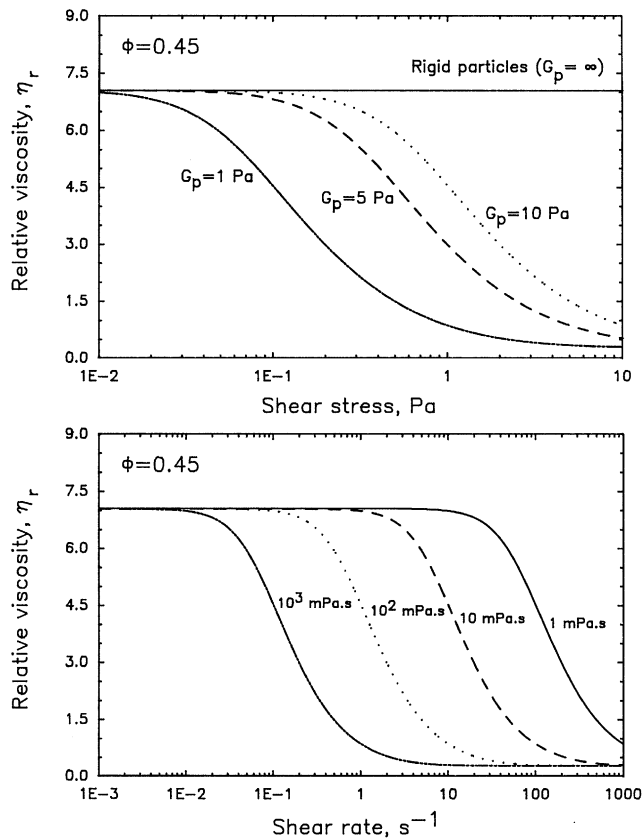


Fig. 3. Effects of particle rigidity and continuous medium viscosity (η_c) on the relative viscosity of a concentrated suspension. The plots are generated from model 4 (Eq. (18)) using $[\eta] = 2.5$ and $\phi_m = 0.637$. G_p is the particle shear modulus.

saline. Rheological measurements were made at 25°C using a Weissenberg rheogoniometer.

Chien et al. (1970) as well as Brooks et al. (1970) found that hardened RBC suspensions behave as nearly Newtonian fluids over a wide range of RBC volume fraction ($0 \leq \phi \leq 0.522$). The suspensions of normal RBCs, however, exhibited shear-thinning behavior. As there was no aggregation of cells, and the normal and hardened erythrocytes differed mainly in their respective rigidities, the shear-thinning observed in normal RBC suspensions was due to cellular deformation.

Fig. 4 shows the comparison between the experimental data of hardened RBC suspensions and predictions of various models developed in the paper. As can be seen, the experimental data of both Chien et al. (1970) and Brooks et al. (1970) can be described very well with model 4 (Eq. (18)) using $[\eta] = 3.83$ and $\phi_m = 0.60$. Model 3 (Eq. (14)) overpredicts the relative viscosity whereas models 2 (Eq. (10)) and 1 (Eq. (6)) underpredict η_r . Note that $[\eta]$ and ϕ_m values of 3.83 and 0.60, respectively are quite reasonable. Only for spherical particles, $[\eta] = 2.5$; for non-spherical particles such as RBCs, $[\eta]$ is expected to be greater than 2.5. A ϕ_m value of 0.60 corresponds to hexagonally packed sheets arrangement of monodisperse spheres.

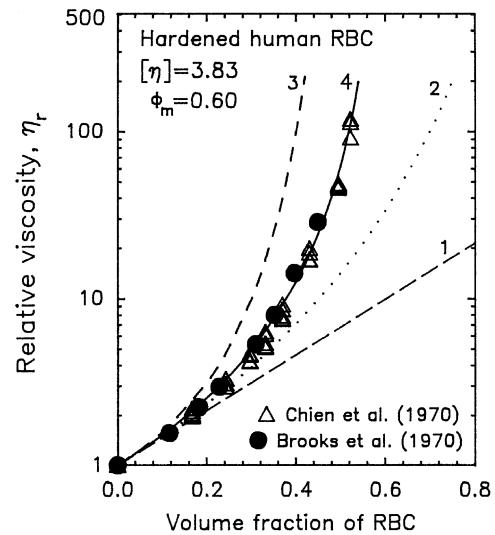


Fig. 4. Relative viscosity (η_r) as a function of volume fraction of hardened RBCs. Comparison between experimental data and predictions of various models. Numbers shown on different curves refer to model number.

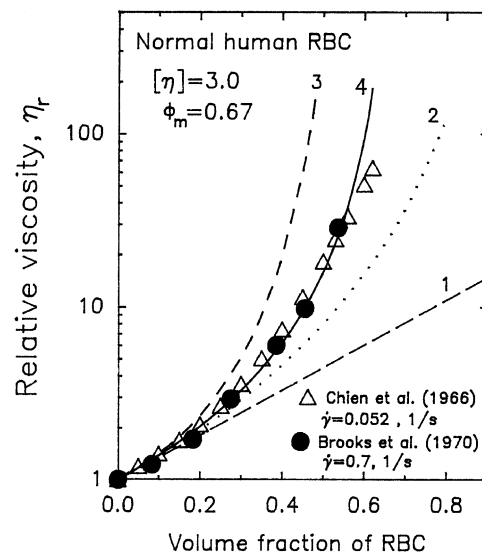


Fig. 5. Relative viscosity (η_r) as a function of volume fraction of normal RBCs at low shear rates. Comparison between experimental data and predictions of various models. Numbers shown on different curves refer to model number.

Fig. 5 shows the comparison between the experimental data of normal RBC suspensions at low shear rate and predictions of various models. Again, model 4 (Eq. (18)) describes the experimental data very well when $[\eta]$ and ϕ_m values of 3.0 and 0.67, respectively, are used. As in the case of hardened RBC suspensions, model 3 overpredicts η_r whereas models 2 and 1 underpredict η_r .

As mentioned earlier, suspensions of normal RBCs exhibit shear-thinning behavior. Figs. 6 and 7 show the experimental η_r data of Chien et al. (1966) and Brooks et al. (1970) as a function of shear rate for the normal

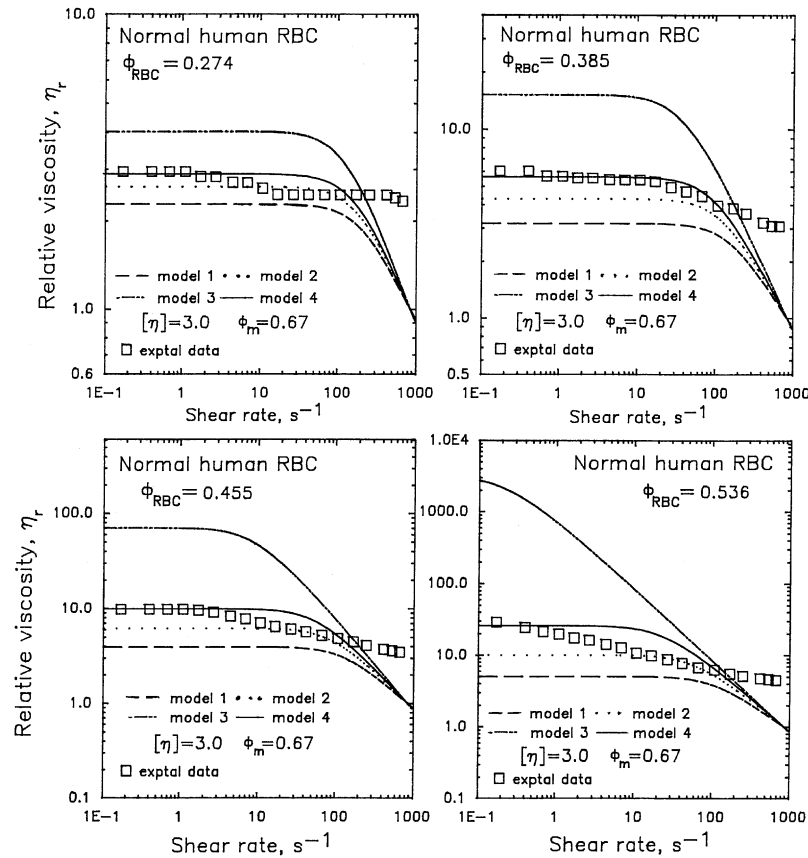


Fig. 6. Relative viscosity (η_r) as a function of shear rate. Comparison between experimental data of Brooks et al. (1970) on normal RBC suspensions and predictions of various models.

RBC suspensions. The predictions of various models developed in the paper are also shown. The $[\eta]$ and ϕ_m values of 3.0 and 0.67, respectively, determined earlier from low shear data of normal RBC suspensions, were used in the model calculations. The shear modulus of particles (RBC) was taken to be 1 Pa, that is, $G_p = 1$ Pa. This value of G_p corresponds to the ratio of elastic shear modulus of the RBC membrane to the average radius of the RBC. Several authors in the literature (Skalak et al., 1989; Breyiannis and Pozrikidis, 2000) have reported that the elastic shear modulus of the RBC membrane is about 4×10^{-6} N/m and the average radius of the RBC is $4 \mu\text{m}$. The experimental data of Brooks et al. show reasonably good agreement with model 4 (Eq. (18)) predictions when shear rates are less than 200 s^{-1} (Fig. 5). This shear rate of 200 s^{-1} corresponds to a shear modulus number (N_{se}) of about 0.2. There are two possible causes for the observed deviation between experimental data and model predictions at high shear rates: first, at high shear rates, the motion of RBCs may change from flipping/tumbling type to tank-treading type in which the cell deforms into an ellipsoid-like particle at a stationary orientation with the membrane rotating about the cell interior; and second, the Goddard and Miller model (Eq. (1)), and hence the models

developed in the paper, are valid only for $N_{se} < 1.0$. The experimental data of Chien et al. also show reasonably good agreement with model 4 (Eq. (18)) predictions, especially for $\phi \leq 0.45$ (Fig. 7). At higher ϕ , model 2 (Eq. (10)) appears to give better predictions. From the data shown in Figs. 6 and 7, it is clear that model 3 (Eq. (14)) severely overpredicts the relative viscosities of RBC suspensions; model 1 (Eq. (6)) on the other hand underpredicts the η_r values. The η_r values predicted by model 2 (Eq. (10)) are generally close to the predictions of model 4 (Eq. (18)).

5. Concluding remarks

Starting from the viscosity equation for a dilute suspension of deformable elastic particles, four new equations have been developed for the viscosity of concentrated suspensions of deformable elastic particles, using the DEMA. Out of the four models developed in the paper, two models predict the relative viscosity of suspensions to be a function of two variables: shear modulus number, defined as the ratio of shear stress to elastic shear modulus of particle, and volume fraction of particles. The remaining two models include an

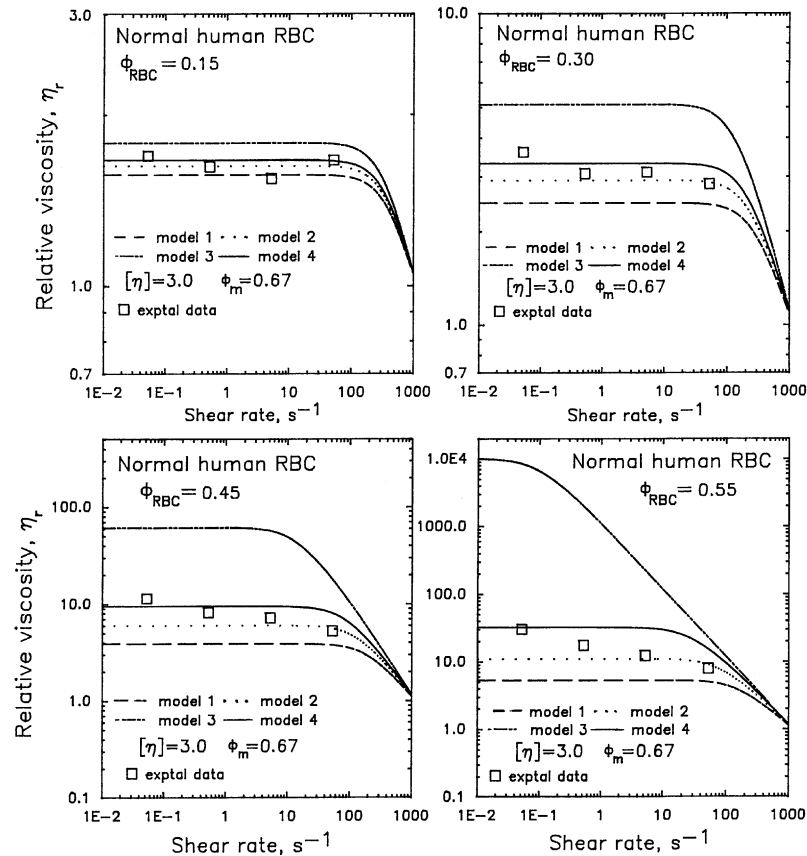


Fig. 7. Relative viscosity (η_r) as a function of shear rate. Comparison between experimental data of Chien et al. (1966) on normal RBC suspensions and predictions of various models.

additional parameter, that is, the maximum packing volume fraction of particles. The proposed models are capable of describing the rheological behavior of un-aggregated suspensions of human erythrocytes. A decrease in relative viscosity and an enhancement of shear-thinning behavior is predicted when either the particle rigidity is decreased or the continuous phase viscosity is increased. These predictions are similar to those observed in the case of un-aggregated suspensions of RBCs. The proposed models are evaluated using the literature data on relative viscosity of normal and hardened human RBC suspensions in protein-free saline.

Acknowledgements

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